

Zhang Yangruo

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Address: Zhuhai, Guangdong, China

EDUCATION

Beijing Normal-Hong Kong Baptist University 09/2023 – Present
Bachelor of Science (Honours) in Applied Mathematics GPA: 3.41/4.0
Minor in Computer Science Expected Graduation: Jun. 2027
Main Coursework: Probability Theory, Mathematical Statistics, Applied Stochastic Process, Optimization, Numerical Analysis, Mathematical Modelling, Machine Learning, Data Structures and Algorithms, Database Management Systems, Graph Theory, Discrete Mathematics

INTERNSHIP EXPERIENCE

Chia Tai Tianqing Pharmaceutical Group Co., Ltd. Jiangsu, China
Biopharmaceutical Development Research Center – Biologics R&D Assistant Jun. 2026 – Sep. 2026

- Conducted literature review and biomedical data organization to support biologics research and development projects.
- Performed data cleaning, descriptive statistical analysis, and basic data visualization for biomedical research datasets.
- Assisted with research support tasks by maintaining organized documentation and improving data readability for subsequent analysis.
- Developed practical exposure to pharmaceutical R&D workflows, biomedical data processing, and evidence-based research support.

Ma'anshan Tianyuan Surveying and Mapping Co., Ltd. Anhui, China
Data Analysis Department – Remote Sensing Data Analyst Intern Jan. 2026 – Mar. 2026

- Validated satellite-based inversion models for vegetation-cover and water-quality indicators using regression analysis, hypothesis testing, and statistical reliability assessment.
- Applied Monte Carlo simulations to quantify uncertainty propagation in remote-sensing-derived parameter estimates.
- Conducted time-series analysis of seasonal vegetation-index patterns to support environmental monitoring and model validation.

Jiangsu Hoperun Software Co., Ltd. Jiangsu, China
Huawei Enterprise Department – Assistant Data Annotator Jun. 2025 – Aug. 2025

- Processed and annotated enterprise datasets for Huawei-related machine learning workflows, supporting the construction of high-quality ground-truth data.
- Performed data quality assurance by identifying anomalies, resolving edge cases, and improving consistency in unstructured data.

RESEARCH PROJECTS

ADAPT: Forecast-Guided Adaptive Diagnostic Triage for Dengue Jan. 2026 – Apr. 2026
First Author; Submitted to Computer Methods and Programs in Biomedicine, Apr. 2026

- Developed ADAPT, a forecast-guided decision-support framework linking weekly dengue RT-PCR-confirmed positivity forecasting with cost-sensitive PCR triage.
- Processed 78,068 dual-tested Brazilian SINAN dengue surveillance records from 2021–2025 and constructed a reproducible weekly positivity time series.
- Implemented a bounded-output attention-guided LSTM in PyTorch, achieving held-out test MSE of 0.091, MAE of 0.227, and Pearson correlation of 0.719.
- Derived a closed-form analytical decision threshold and used Monte Carlo simulations to evaluate diagnostic-policy robustness under stochastic sensitivity and specificity assumptions.
- In an 18-week held-out policy evaluation, ADAPT improved PPV from 0.907 to 0.917 and reduced expected false positives from 12.99 to 11.43 relative to static NS1 screening, while remaining substantially less costly than universal PCR confirmation.

TME Hypergraph-based Perturbation Response Prediction

Sep. 2025 – Jan. 2026

Third Author; Submitted to Bioinformatics, May. 2026

- Addressed the limitation of existing single-cell perturbation-response models in capturing higher-order spatial organization and local niche signals within the tumor microenvironment.
- Proposed a semantic hypergraph-based framework to encode biologically motivated microenvironmental motifs and structurally improved the chemCPA baseline model for rigorous benchmarking.
- Synthesized benchmark evaluation results showing strong predictive agreement across multiple datasets; contributed to manuscript writing and model evaluation as third author.

HONORS AND ACADEMIC ACTIVITIES

China Undergraduate Mathematical Contest in Modeling – Provincial Second Prize	Oct. 2025
China Undergraduate Mathematical Contest in Modeling – Provincial Third Prize	Oct. 2024
Imperial College London – Certificate in Statistics & Data Analysis in Public Health	Jan. 2026
Johns Hopkins University – Biostatistics in Public Health Specialization	Apr. 2026 – May. 2026
Leader Path Programme of BNBU – Gold Award	May. 2026
Leader Path Programme of BNBU – Silver Award	May. 2025
The 14th National Forum on Mathematical Culture – Conference Attendee	Jun. 2025 – Jul. 2025

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

• Executive Committee Member , BNBU Student Union	Sept. 2023 – Jun. 2025
• Student Administrative Assistant , BNBU Admission Office	May. 2024 – Jul. 2024
• Peer Mentor , BNBU Peer Mentor Program	Sept. 2024 – Jun. 2025
• Chairperson of Electoral Commission , BNBU Student Union	Sept. 2025 – Jun. 2026
• Head of Organization Department , BNBU CCYL Committee	Sept. 2025 – Jun. 2026
• Resident Tutor , BNBU Student Affairs Office	Sept. 2025 – Jun. 2026
• Participant , Guangdong province "Qingma Project" Youth Leadership Training Program	Jun. 2026

SKILLS

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- **Programming & Tools:** Python, PyTorch, MATLAB, C/C++, MySQL, Linux, Matplotlib, Seaborn
 - **Statistical & Machine Learning Methods:** Bayesian Inference, Hypothesis Testing, Monte Carlo Simulation, Time-Series Forecasting, LSTM, Attention-based Models, Graph Neural Networks, Decision Optimization, Uncertainty Quantification
 - **Languages:** Mandarin Chinese (Native), English (IELTS Academic 6.5)